

Breast Cancer Diagnosis Classification

An End-to-End Supervised Machine Learning Pipeline

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This report documents an educational coursework project. The model described is not a medical device and is not suitable for clinical use.

1. Executive Summary

This project develops and evaluates a supervised machine learning pipeline that classifies breast tumours as benign or malignant using clinical and pathological features from 213 patient records collected at the University of Calabar Teaching Hospital between January 2019 and August 2021.

Five classification algorithms were compared: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbours. Each was tuned through grid search and validated using repeated stratified cross-validation. The Decision Tree achieved the highest cross-validated F1 score, and nested cross-validation produced an unbiased performance estimate of 0.840 ± 0.036 F1.

On the held-out test set of 43 patients, the final model reached 90.7% accuracy, 94.1% precision, 84.2% recall, and a ROC-AUC of 0.958. The model was deployed as an interactive web application using Gradio.

Two findings carry more analytical weight than the headline metrics. First, all five algorithms performed within statistical noise of each other, so model selection was not the decisive factor. Second, the decision threshold, not the choice of algorithm, controls the trade-off that actually matters in a diagnostic setting, where a missed malignancy costs far more than a false alarm.

2. Problem Definition and Dataset

2.1 Problem statement

Given routinely collected clinical and pathological measurements, can a machine learning model reliably distinguish malignant from benign breast tumors? The practical framing is a decision-support tool: a model that flags cases warranting closer review, with an operating point deliberately tuned to minimize missed cancers.

2.2 Dataset description

Attribute	Value
Source	Kaggle — Breast Cancer Analysis
Origin	University of Calabar Teaching Hospital Cancer Registry
Collection period	January 2019 – August 2021 (24 months)
Raw records	213
Records after cleaning	211
Raw features	11
Predictors used	8
Target variable	Diagnosis Result (Benign / Malignant)
Class distribution	Benign 119 (56.4%) · Malignant 92 (43.6%)

The class balance is close to even, which means accuracy is not misleading in the way it would be on a heavily imbalanced dataset. Nonetheless F1 was used as the primary optimization metric throughout, because it weights performance on the malignant class directly.

2.3 Feature definitions

Feature	Type	Description
Age	Numeric	Patient age at diagnosis (range 13–77)
Menopause	Binary	Menopausal status at time of diagnosis
Tumor Size (cm)	Numeric	Size of the excised tumor
Inv-Nodes	Numeric	Axillary lymph nodes containing metastatic tissue
Breast	Categorical	Affected side — Left or Right
Metastasis	Binary	Cancer present in other organs
Breast Quadrant	Categorical	Upper/Lower × Inner/Outer section of the gland
History	Binary	Family history of breast cancer

Two columns were removed before modelling. S/N is a patient identifier and Year of Diagnosis is an administrative field; neither carries clinical signal, and retaining an identifier is a common source of spurious model performance.

3. Data Preprocessing and Exploration

3.1 Missing data

An initial null check returned zero missing values across all columns. This result was misleading. The raw file uses the character "#" as a placeholder for unrecorded data, which pandas read as a valid string rather than a null. Identifying this required inspecting the value counts of each column rather than trusting the null count.

Handling differed by column type:

- Categorical columns, values were stripped and title-cased, then "#" was mapped to an explicit "Unknown" level. This preserves the information that a value was not recorded, rather than discarding those rows.
- Numeric columns, "#" was coerced to NaN using `pd.to_numeric(errors="coerce")`, allowing the columns to convert to float.

After conversion, five rows contained genuine nulls and were dropped, reducing the dataset from 213 to 211 records. Zero duplicate rows were present.

3.2 Preprocessing pipeline

All transformations were wrapped in a scikit-learn `ColumnTransformer` nested inside a `Pipeline`. This is a deliberate methodological choice: it guarantees that imputation statistics, scaling parameters, and encoder categories are computed only from the training portion of each cross-validation fold.

Column type	Transformation sequence
Numeric	<code>SimpleImputer(strategy="median")</code> → <code>StandardScaler</code>
Categorical	<code>SimpleImputer(strategy="most_frequent")</code> → <code>OneHotEncoder(handle_unknown="ignore")</code>

Fitting a scaler or imputer on the full dataset before splitting is one of the most common sources of data leakage in student projects. It inflates validation scores because information from the held-out fold has influenced the transformation. Placing every step inside the pipeline eliminates this by construction.

The `handle_unknown="ignore"` setting on the encoder ensures that a category appearing only in the test fold does not raise an error at prediction time.

3.3 Exploratory findings

Boxplots of the numeric predictors by diagnosis show clear separation for tumor size: malignant cases center around 6 cm against roughly 3 cm for benign cases. Involved nodes and metastasis show near-complete separation, with benign cases concentrated at zero.

The correlation matrix reveals strong relationships among the clinical severity indicators:

Variable pair	Correlation
Metastasis ↔ Inv-Nodes	0.88
Tumor Size ↔ Inv-Nodes	0.72
Tumor Size ↔ Metastasis	0.70
Age ↔ Menopause	-0.71

The high inter-correlation among tumor size, involved nodes, and metastasis is clinically expected, these measurements describe the same underlying disease progression. For modelling this means the effective information content of the feature set is lower than the column count suggests, which helps explain why feature selection later produced no improvement.

The Age–Menopause correlation of -0.71 is an artefact of how menopause was encoded in this dataset (0 = with menopause, 1 = no menopause yet), so the negative sign reflects the coding rather than a counterintuitive relationship.

3.4 Train/test split

The data were split 80/20 with stratification on the target, producing 168 training and 43 test records. Class balance held at 56.5/43.5 in training against 55.8/44.2 in test, close enough that evaluation metrics are not skewed by an unrepresentative split.

4. Model Selection and Development

4.1 Baseline comparison

Five algorithms were fitted with default hyperparameters and evaluated on the held-out test set:

Model	Accuracy	Malignant recall	Missed cancers
Decision Tree	0.930	0.84	3
Random Forest	0.930	0.84	3
Logistic Regression	0.907	0.79	4
SVM	0.884	0.74	5
KNN	0.860	0.68	6

A pattern worth noting: every model produced zero false positives, giving all five a malignant precision of exactly 1.00. Every error was a missed cancer. This means the accuracy ranking is driven entirely by recall, and accuracy adds no information beyond the count of false negatives.

The differences are also not statistically meaningful. With 43 test records, a single patient shifts accuracy by 2.3 percentage points. The full spread from best to worst is three patients, and the Decision Tree and Random Forest produced identical confusion matrices.

4.2 Hyperparameter tuning

Each model was tuned with GridSearchCV using 5-fold stratified cross-validation and F1 scoring. Grid sizes ranged from 20 configurations for Logistic Regression to 60 for KNN.

Model	CV F1	Train F1	Gap
Decision Tree	0.894	0.911	0.017
Logistic Regression	0.874	0.881	0.007
SVM	0.874	0.879	0.005
KNN	0.874	0.881	0.007
Random Forest	0.868	0.935	0.067

The train–validation gap column is diagnostic. Random Forest shows the widest gap at 0.067, indicating it memorizes the training folds more than the others without translating that into validation performance, expected behavior for a high-capacity ensemble on 168 rows.

Random search was also applied over continuous distributions for Logistic Regression, Random Forest, and SVM with a budget of 50 iterations each. Random search sampled a Random Forest configuration

scoring 0.877, slightly above the grid search result, illustrating that continuous hyperparameters such as C and gamma are not well served by a fixed lattice of candidate values.

4.3 Robustness assessment

Single cross-validation on a small dataset is itself noisy. Repeated stratified cross-validation (5 splits $\times$ 10 repeats = 50 folds) was used to obtain a stable estimate for each tuned model:

Model	F1 (mean $\pm$ std)	Minimum	Maximum
Decision Tree	0.877 $\pm$ 0.062	0.720	0.966
SVM	0.876 $\pm$ 0.065	0.750	1.000
KNN	0.875 $\pm$ 0.064	0.750	1.000
Logistic Regression	0.873 $\pm$ 0.065	0.750	1.000
Random Forest	0.862 $\pm$ 0.064	0.741	0.966

The conclusion here is more useful than a ranking: all five models are statistically indistinguishable. The total spread of 0.015 sits well inside a standard deviation of approximately 0.064. Selecting the Decision Tree as the final model is defensible on parsimony and interpretability grounds, but claiming it outperforms the alternatives would not be supported by the evidence.

4.4 Nested cross-validation

Standard cross-validation reports an optimistically biased score when hyperparameters have been tuned on the same folds used for scoring. Nested cross-validation corrects this by placing the entire grid search inside an inner loop.

Estimate	F1	Std
Standard repeated CV	0.877	0.062
Nested CV	0.840	0.036

The 0.037 gap quantifies the optimism introduced by tuning. The nested figure of 0.840 $\pm$ 0.036 is the honest estimate of how the modelling procedure would perform on new data, and it is the number reported in the deployed application. The reduced standard deviation reflects averaging over a larger number of fold combinations.

5. Model Evaluation and Optimization

5.1 Final test set performance

Metric	Value
Accuracy	0.907
Precision (Malignant)	0.941
Recall (Malignant)	0.842
F1 (Malignant)	0.889
ROC-AUC	0.958

Confusion matrix: 23 true negatives, 1 false positive, 3 false negatives, 16 true positives.

The three false negatives are the errors that matter. Each represents a malignant case classified as benign. The single false positive represents a benign case sent for unnecessary follow-up. These two error types are not equivalent in cost, and the evaluation should not treat them as such.

The ROC curve shows an area of 0.958 and the precision-recall curve an average precision of 0.940, indicating strong separation between classes. Both curves display an angular, step-like shape rather than a smooth arc, a consequence of the tree producing discrete probability values rather than continuous ones.

5.2 Threshold analysis

Attempting to tune the decision threshold on the Decision Tree revealed a structural limitation. The tuned tree has only four leaves, so it emits exactly four distinct probability values: 0.00, 0.19, 0.64, and 1.00. Any threshold between 0.20 and 0.60 produces identical predictions.

A sweep from 0.05 to 0.95 in 0.05 increments produced only three distinct outcomes, and the only threshold that changes anything below 0.19 would reclassify thirteen cases simultaneously. The threshold is effectively uncontrollable on this model.

Logistic Regression was therefore fitted as a complementary model to expose the trade-off, since it produces continuous probability:

Threshold	Recall	Precision	Missed cancers	False alarms
0.2	0.842	0.889	3	2
0.3	0.842	0.941	3	1
0.4	0.789	1.000	4	0
0.5 (default)	0.789	1.000	4	0

A threshold of 0.3 is the sensible operating point. It achieves the same recall as 0.2 with one fewer false alarm, so lowering further buys nothing. Moving up to the conventional 0.5 gives up one malignant case in exchange for eliminating one false positive, the wrong direction for a diagnostic screen, since missing a cancer is not equivalent to sending a patient for an unnecessary follow-up.

The broader point is that 0.5 is a convention inherited from the default behavior of classification libraries, not an optimum. Selecting a threshold based on the relative cost of the two error types is a modelling decision that should be made explicitly and documented.

5.3 Feature importance

Permutation importance was computed on the test set over 30 repeats, measuring the drop in F1 when each column is shuffled:

Feature	Importance
Tumor Size (cm)	0.127
Inv-Nodes	0.126
Menopause	0.069
Breast Quadrant	0.048
Age	0.006
Breast, Metastasis, History	0.000

Tumor size and involved nodes dominate, consistent with the exploratory correlation analysis. Permutation importance was chosen over the tree's built-in `feature_importances_` because it operates on the raw input columns regardless of downstream one-hot encoding, making the results directly interpretable in terms of the original variables.

The zero importance for Metastasis is notable given its 0.88 correlation with Inv-Nodes. Because the two carry nearly the same information, shuffling one leaves the model able to recover the signal from the other, a known characteristic of permutation importance under correlated features, not evidence that metastasis is clinically irrelevant.

5.4 Feature selection

Recursive Feature Elimination with cross-validation (RFECV) was placed inside the pipeline so that selection is refit on each training fold. Selecting features on the full dataset before cross-validation is a common leakage mistake that inflates reported performance.

Configuration	F1 (50 folds)
All features (grid search)	0.877 $\pm$ 0.062
RFECV subset	0.858 $\pm$ 0.062

RFECV retained all nine encoded features and produced no improvement. The 0.019 difference is well within one standard deviation, so the honest interpretation is that feature selection made no difference on this dataset. Given only eight predictors, of which four carry meaningful signals, there was limited scope for elimination to help.

5.5 Regularization

Regularization was assessed on Logistic Regression across 50 cross-validation folds:

Penalty	F1 (mean $\pm$ std)
No penalty	0.841 $\pm$ 0.063
L2 (Ridge)	0.859 $\pm$ 0.061
L1 (Lasso)	0.860 $\pm$ 0.061
ElasticNet	0.859 $\pm$ 0.060

All three penalties outperform the unpenalized model by roughly 0.018, and the three are indistinguishable from each other. The magnitude is a weak signal relative to the 0.06 standard deviation, but the direction is consistent and matches expectation for 168 training rows.

Tuning the regularization strength selected $C = 0.1$ with an L1 penalty, scoring 0.868. The regularization path is more informative than the single best value: as C increases from 0.001 to 100, training F1 climbs from 0.872 to 0.905 while validation F1 falls from 0.864 to 0.829. The widening gap is a textbook demonstration of overfitting as the penalty weakens and confirms that stronger regularization is appropriate at this sample size.

The selected L1 model drove 10 of 14 encoded coefficients to exactly zero, retaining only Inv-Nodes, Tumor Size, Metastasis, and Age. This independently corroborates the permutation importance ranking.

Because Decision Trees are not regularized through L1 or L2 penalties, the equivalent constraints were tested separately:

Tree constraint	F1 (mean $\pm$ std)
Unconstrained	0.820 $\pm$ 0.053
Depth-limited (max_depth = 4)	0.846 $\pm$ 0.065
Leaf-constrained (min_samples_leaf = 8)	0.850 $\pm$ 0.064

Tree constraint	F1 (mean $\pm$ std)
Cost-complexity pruning (ccp_alpha = 0.01)	0.849 $\pm$ 0.068

All three constrained variants outperform the unconstrained tree, confirming that complexity control improves generalization for this model family as well.

5.6 Learning curve

The learning curve shows training and validation scores converging with no divergence, indicating the final model is not overfitting. Validation F1 rises from 0.80 at 34 training examples to 0.89 at 135 and is still climbing at the largest training size tested. The practical implication is that additional data would improve performance, the model is limited by sample size rather than by capacity.

6. Deployment

6.1 Model artifact

The complete fitted pipeline was serialized with joblib as a dictionary containing three elements: the pipeline itself, the decision threshold, and the ordered list of feature names. Bundling the feature list guards against column-order drift at inference time, which would otherwise produce silently incorrect predictions rather than an error.

6.2 Interface

An interactive web application was built with Gradio, structured in three tabs:

- Single patient = sliders and dropdowns for each predictor, returning a colour-coded verdict with the probability of malignancy, a visual confidence bar, and the active decision threshold.
- Batch scoring = CSV upload with column validation, returning scored rows and a downloadable predictions file.
- About= dataset provenance, methodology summary, the nested cross-validation figure, and stated limitations.

Dropdown values for the categorical inputs were pulled directly from the fitted encoder's categories rather than typed manually, ensuring the interface cannot submit a value the model has not seen.

The application was deployed to Hugging Face Spaces for permanent hosting, replacing the temporary public URL that Gradio generates from within Colab.

The screenshot shows a web application interface with two main sections: 'Patient' and 'Tumour'. The 'Patient' section includes sliders for 'Age' (set to 44) and 'Menopause (0 = pre, 1 = post)' (set to 0), and a radio button for 'Family history' (set to 1). The 'Tumour' section includes a slider for 'Tumour size (cm)' (set to 0.5). On the right, a green-bordered box displays the prediction: 'Benign' in green text, followed by 'Probability of malignancy: 20.0%' and 'Certainty: Moderate'. A green progress bar indicates the probability level. Below this, it states 'Decision threshold: 30% — set below 50% to prioritise sensitivity'. A yellow box at the bottom contains a disclaimer: 'Educational coursework demo. Not a medical device and not a substitute for clinical diagnosis.'

Single patient

Batch (CSV)

Model

Patient

Age

23

↕

20

90

Menopause (0 = pre, 1 = post)

0

1

Family history

0

1

Tumour

Malignant

Probability of malignancy: **100.0%** | Certainty: High

Decision threshold: 30% — set below 50% to prioritise sensitivity

Educational coursework demo. Not a medical device and not a substitute for clinical diagnosis.

Public URL: <https://huggingface.co/spaces/AyleenSantandere/breast-cancer-classifier>

Spaces

AyleenSantandere

breast-cancer-classifier

Like 0

Agents

Paused

Logs

App

Files

Community

Settings

Space Paused

You've reached your CPU Basic quota limit across all your Spaces!

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Restart this Space

Hugging Face

Search models, datasets, users...

Models

Datasets

Spaces

Buckets **NEW**

Docs

Enterprise

Pricing

Spaces: AyleenSantandere

breast-cancer-classifier

Like 0

Running

App

Files

Community

Settings

Want to edit your Spaces's metadata? Head to the [README.md and metadata UI](#) instead.

Space Access control

Public

This Space is currently `public`. Anyone on the internet can see this Space. Only you can commit.

Git Storage Usage

Storage overview

List Large Files

Large File Storage Used

5.71 kB

Git Storage limit

1 GB

Variables and secrets

New variable

New secret

Variables

Public

No variables

Secrets

Private

Secrets are not available for static Spaces

Custom domain

Submit

13

7. Conclusions

The project delivers a working classification pipeline with an honest generalization estimate of 0.840 ± 0.036 F1 and test set performance of 90.7% accuracy with 84.2% recall on the malignant class. The pipeline architecture prevents data leakage by construction, and the evaluation methodology, repeated cross-validation followed by nested cross-validation, is appropriate for a dataset of this size.

Three conclusions are worth carrying forward.

Algorithm choice mattered less than expected. All five candidate models scored within noise of one another across 50 cross-validation folds. Effort spent selecting among them yielded no measurable benefit, while the threshold decision changed outcomes for real patients in the test set.

Probability granularity is a model property worth checking early. The Decision Tree's four-leaf structure made threshold tuning impossible, which was not apparent from its accuracy or F1 score. This only surfaced when the threshold sweep returned identical results across a wide range of values.

Sample size is the binding constraint. With 43 test records, differences of fewer than three patients are indistinguishable from noise, and the learning curve indicates performance had not plateaued. The most effective improvement to this project would be more data, not a more sophisticated algorithm.

8. Limitations

- Sample size = 211 records total with 43 in the test set. One patient shifts test accuracy by 2.3 percentage points, so model rankings derived from the test set are not statistically meaningful.
- Single source = data originates from one hospital registry over a 24-month window. No external validation cohort was available, so generalization to other populations is untested.
- Probability calibration = predicted probabilities were not calibrated. They should be interpreted as ranking scores rather than as true risk estimates.
- Feature scope = the dataset contains no imaging, histological grade, or receptor status, all of which are standard inputs to clinical diagnosis. The feature set represents a small subset of available diagnostic information.
- Placeholder handling = five records were dropped due to unrecorded values. With a dataset this small, the removal of even a few records affects results.
- Not clinically validated = this is a coursework demonstration of an end-to-end machine learning workflow. It has not undergone clinical evaluation and is not suitable for use in patient care.

Appendix: Technical Stack

Category	Tools
Language	Python 3
Data handling	pandas, NumPy
Modelling	scikit-learn 1.6.1
Visualization	matplotlib, seaborn
Serialization	joblib 1.6.0
Interface	Gradio
Environment	Google Colab
Hosting	Hugging Face Spaces

Key scikit-learn components: Pipeline, ColumnTransformer, SimpleImputer, StandardScaler, OneHotEncoder, GridSearchCV, RandomizedSearchCV, StratifiedKFold, RepeatedStratifiedKFold, RFECV, permutation_importance, learning_curve.